

rule that relies on percentages rather than hard-dollar amounts. With their framework, the ceiling refers to a maximum *percentage* increase in spending each year, while the floor refers to a maximum *percentage* drop in spending for each year. The baseline case they describe is a 5 percent ceiling and a 2.5 percent floor, illustrated in Exhibit 6.10.

To determine spending, they say to first apply a fixed-percentage spending rule to the remaining account balance. For the baseline, if the spending amount would be more than 5 percent greater than that of the previous year, then the new spending amount is capped at 5 percent growth. Meanwhile, if the spending amount would drop by more than 2.5 percent, then spending for the year is limited to a 2.5 percent decrease. Each year the rule is applied again to the previous year's spending level.

The narrower the range between the floor and ceiling, the more this spending rule acts as a constant amount rule with a fixed spending growth rate. As the range increases, the rule behaves more like a fixed-percentage rule, because you become less likely to reach the binding constraints on either end. Despite the lack of a hard-dollar floor, this method is also susceptible to portfolio depletion. Spending may not always drop quickly enough to preserve the portfolio.

Exhibit 6.10

Time Path of Real Spending and Wealth

Vanguard's Percentage Floor-and-Ceiling Rule

For 4% Initial Spending Rate, 50/50 Asset Allocation, Rolling 30-Year Retirements

Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds